

Last time

$A \subseteq \mathbb{R}$ is dense if for all $I \subseteq \mathbb{R}$ open interval, $A \cap I \neq \emptyset$

- Showed that $\mathbb{Q}$ are dense in $\mathbb{R}$.
- Are irrationals dense in $\mathbb{R}$?

yes: Let $(x, y) \subseteq \mathbb{R}$

- If x, y were in $\mathbb{Q}$



Scale by
 $y-x$



Scale by $y-x$



irr. b/c $\sqrt{2}$ is irrational

So now if we could find $(x', y') \subseteq (x, y)$, where $x', y' \in \mathbb{Q}$
b/c (x', y') contains an irr. # by previous argument.

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, $\exists x' \in (x, y) \cap \mathbb{Q}$

Now look at the open interval $(x', y) \subseteq (x, y)$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, $(x', y) \cap \mathbb{Q} \ni y'$ exists

so $(x', y') \subseteq (x, y)$



Uniform Continuity

Def. The function f is uniformly continuous on an interval $I \subseteq \mathbb{R}$ if $\forall \varepsilon > 0, \exists \delta > 0$ s.t.

whenever $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

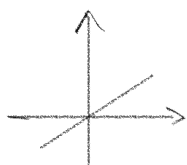
- usual continuity (at a pt a , w/ limit L)

Given a $\varepsilon \rightarrow$ find δ , but δ depends on a .

- uniform continuity (for any point y)

Given an $\varepsilon \rightarrow$ same δ for any pt in I .

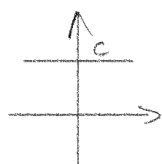
Ex ① $y = x$



Given $\varepsilon > 0$ at any pt.

We've seen that $\delta = \varepsilon$ works.

② $y = c$



Given $\varepsilon > 0$, any $\delta > 0$ works

③ $y = \frac{1}{x}$ is not uniformly cts.

If it were unif. cts, then for $\varepsilon = 1$, $\exists \delta > 0$ s.t. $\forall |x - y| < \delta, |f(x) - f(y)| < \varepsilon$

Let $n \geq 1$ be an integer

$$x = \frac{\delta}{2n}, \quad y = \frac{2\delta}{2n}$$

$$f(x) = \frac{2n}{\delta}, \quad f(y) = \frac{n}{\delta} \Rightarrow |f(x) - f(y)| = \frac{n}{\delta}$$

as $n \rightarrow \infty$, $|f(x) - f(y)| \rightarrow \infty$

(In fact, if $n > \delta$, then $|f(x) - f(y)| > 1 = \varepsilon$.)

So not unif. cts.

(4) $y = x^2$ is not unif. cts.

If it were unif. cts., then for $\varepsilon = 1$, $\exists \delta > 0$ s.t.

Whenever $|x - y| < \delta$, $|f(x) - f(y)| < 1$.

Let $x = n$, $y = n + \frac{\delta}{2}$

$$\begin{aligned} |f(x) - f(y)| &= |n^2 - (n^2 + n\delta + \frac{\delta^2}{4})| \\ &= |n\delta + \frac{\delta^2}{4}| \end{aligned}$$

By Δ Ineq, $|n\delta + \frac{\delta^2}{4}| + |-\frac{\delta^2}{4}| \geq |n\delta|$

$$|n\delta + \frac{\delta^2}{4}| \rightarrow \infty \text{ as } n \rightarrow \infty$$

So for n large ($n > \frac{1}{\delta}$), $|n\delta + \frac{\delta^2}{4}| > |1 + \frac{\delta^2}{4}| > 1 = \varepsilon$

So $f(x) = x^2$ is not uniformly continuous.

Thm. If f is cts. on $[a, b]$, then f is unif. cts. on $[a, b]$.

Pf. Want to find $\delta > 0$, s.t.

whenever $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$

$A_\varepsilon = \{y \in [a, b] : \exists \delta > 0 \text{ s.t. whenever } |x - x'| < \delta, |f(x) - f(x')| < \varepsilon, x, x' \in [a, y]\}$.

• $A_\varepsilon \neq \emptyset : a \in A_\varepsilon$

By construction, b is an upper bound

$\xRightarrow{\text{P13}}$ LUB exists, call it α (want $\alpha = b$).

Suppose for a contradiction $d \neq b \Rightarrow d < b$

Since f is cts. at d for our given ε ,

$\exists \delta' > 0$, s.t. whenever $|x - d| < \delta'$, $|f(x) - f(d)| < \varepsilon$



$$d - \delta' < x < d + \delta'$$

Let $\delta'' = \min\{\delta', \delta\}$

Then if $|x - y| < \delta''$, either $x, y \in [a, d)$

or $x, y \in (d - \delta', d + \delta')$

so $|f(x) - f(y)| < \varepsilon$ by continuity at d .

So for $d + \frac{\delta'}{2} \in A_\varepsilon$, contradicts that d was the LUB of A_ε

$$\Rightarrow d = b$$

Still have to show $b \in A_\varepsilon$